

XenEcosys : An ecosystem for communication without network connectivity.

The prompt implicates the scenario where all form of connectivity is destroyed and we are left with only hardware capabilities. Prompt demands for next generation communication platform but when no connectivity is there, then platform needs a complete new architecture not only a platform. Thus we are presenting XenEcosys, a complete communication architecture along with platform considering the scenario where no connectivity is there, we are using Bluetooth and wifi throughout XenEcosys.

XenEcosys is a combination of four major pillars :

1. XenComm : The OS level app which users can use to send, receive messages, store contacts locally and broadcast emergency messages through mesh hopping.

2. XenRelay : The OS level software which mediums (laptops, raspberry pi etc) like bus, train, metro or volunteers use to collect messages from sender and share messages to receiver. It is same as a local bus which has multiple stops in its route and each route has its own hub. It has temporary database storage and less storage capacity compared to XenHub. It gets turned on at stops only, conserving battery efficiency compared to mesh hopping.

3. XenHub : The OS level software which hubs (central place of a particular area) uses to register users, store global contact directory, collect and messages to long distances. It has large database storage compared to XenRelay, it runs 24x7 so everything stays updated. XenHub uses DTN (Delay Tolerant Networking) carriers like intercity trains to share messages to different hub at long distances. It's the major decision maker for long distances.

4. XenRoute : The routing algorithm which chooses the best available path for the message to travel considering travel time as priority. First priority is mesh hoping for short distances, other than mesh it decides to use XenRelay or XenHub.

It checks, in order:

1. Is the receiver directly reachable?
2. If not, is there a mesh path through nearby XenComm devices?
3. If not, is a relay or hub available according to the cached routing data?
4. If not, securely store the message and wait for the next synchronization opportunity.

XenEcosys demonstrates that reliable communication is possible even when traditional network infrastructure is unavailable. By integrating XenComm, XenRelay, XenHub, and the XenRoute routing algorithm, the proposed ecosystem leverages Bluetooth, Wi-Fi, and Delay Tolerant Networking to provide resilient, energy-efficient, and scalable message delivery. This architecture enables emergency coordination, local and long-distance communication, and uninterrupted information exchange during disasters or extreme situations, making XenEcosys a practical foundation for next-generation offline communication systems.